

A Paley–II analogue of the Scarpis–Đoković construction for prime powers $q \equiv 1 \pmod{4}$

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Abstract

A Hadamard matrix of order m is a $\{1, -1\}$ -matrix H with $HH^T = mI_m$. Scarpis (1898) showed that for a prime $p \equiv 3 \pmod{4}$ a Hadamard matrix of order $p + 1$ can be lifted to one of order $p(p + 1)$, and Đoković extended the lift to all prime powers $q \equiv 3 \pmod{4}$, asking for the analogous procedure for $q \equiv 1 \pmod{4}$. That case was settled by Farouk and Wang (2020), who lift an input Hadamard matrix of order $2(q + 1)$, subject to eligibility conditions, to one of order $2q(q + 1)$, verifying orthogonality by row-pair counting arguments in which the key inner products are imported from the Hadamard property of the input matrix. This note contributes a different method, not a new existence result. Both routes are constructive; the difference is procedure versus formula: their lift takes choices — an input matrix and a bijection — that can change the output, while the matrix here is given by a choice-free closed form, one matrix for each q . There is no input matrix: the finite bands are given entrywise in closed form by the quadratic character on $\mathbb{F}_q$ composed with the 2×2 Paley–II replacement blocks, and the Hadamard property is proved by four block-Gram identities computed directly from character sums. The outputs differ as well: at $q = 5$ our matrix is not Hadamard-equivalent to the order-60 example printed by Farouk and Wang. We also make the family’s coverage explicit: every order $2q(q + 1)$ whose odd part is at most 3000 is already recorded in the standard tables, and the smallest order lying beyond that audited range, $q = 81$ (order $13284 = 4 \cdot 41 \cdot 81$), also follows from a classical Turyn product using T -matrices of order 41 and Williamson-type matrices of order $81 = 9^2$. We exhibit the witnessing constructions, identify the smallest unaudited order, and tabulate, for $q \leq 3000$, those orders for which a deliberately restricted elementary screen supplies no witness, emphasising that these are *unaudited* rather than open.

1 Introduction

For a prime power q let $\mathbb{F} = \mathbb{F}_q$ and let $\chi: \mathbb{F} \rightarrow \{0, \pm 1\}$ be the quadratic character, $\chi(0) = 0$, $\chi(x) = 1$ for nonzero squares and -1 otherwise. Paley’s two constructions split on the residue of q : for $q \equiv 3 \pmod{4}$ Paley I gives a Hadamard matrix of order $q + 1$, and for $q \equiv 1 \pmod{4}$ Paley II gives one of order $2(q + 1)$ [4].

Scarpis [5] showed that for a prime $p \equiv 3 \pmod{4}$ a Hadamard matrix of order $p + 1$ lifts to one of order $p(p + 1)$. Đoković [11] replaced the prime by an arbitrary prime power: if $q \equiv 3 \pmod{4}$ is a prime power there is a Hadamard matrix of order $q(q + 1)$; his note closes with the open problem of an analogous procedure for $q \equiv 1 \pmod{4}$.

Because the Paley–II seed has order $2(q + 1)$, the natural Scarpis-type target in the $q \equiv 1 \pmod{4}$ case is

$$q \cdot 2(q + 1) = 2q(q + 1).$$

We construct such a matrix directly from χ . Throughout, $q \equiv 1 \pmod{4}$ is a prime power, so $\chi(-1) = 1$; this is the only place the residue class is used, and it is exactly what fails for $q \equiv 3 \pmod{4}$.

The existence of Hadamard matrices of order $2q(q+1)$ for prime powers $q \equiv 1 \pmod{4}$ is due to Farouk and Wang [1]; the contribution of this note is the method of construction and proof. Farouk and Wang obtain the matrix as the output of a lifting procedure applied to an input Hadamard matrix of order $2(q+1)$ meeting certain eligibility conditions, and their verification proceeds row pair by row pair, by counting arguments in which the key inner-product values are imported from the Hadamard property of the input matrix. Here there is no input matrix: the finite bands are written entrywise in the closed form $\psi(\chi(y-x-t))$, and the Hadamard property follows from four block-Gram identities whose ingredients are computed directly from the quadratic character — the sums (2.1) and $\sum_z \chi(z(z-d)) = -1$, together with the block identity $P + ZR = 0$. The proof is thus self-contained at the level of character sums, and the single place where the residue class enters, $\chi(-1) = 1$, is isolated. A secondary contribution is the coverage analysis of Appendix A, which has no counterpart in [1]: it locates the initial range against audited tables of known Hadamard orders and applies a restricted elementary screen beyond that range.

2 The Paley–II blocks

Put

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and let $\psi: \{0, \pm 1\} \rightarrow \{\pm 1\}^{2 \times 2}$ be $\psi(0) = Z$, $\psi(1) = P$, $\psi(-1) = -P$; these are the Paley–II replacement blocks, and $ZR = -P$. Write $\mathcal{R} = J_q \otimes I_2$, where J_q is the $q \times q$ all-ones matrix. We use repeatedly

$$\sum_{u \in \mathbb{F}} \psi(\chi(u)) = Z, \tag{2.1}$$

which holds because the nonzero squares and non-squares are equinumerous.

3 The construction

For $t \in \mathbb{F}$ let K_t be the $2q \times 2q$ matrix with rows and columns indexed by pairs (x, a) , $x \in \mathbb{F}$, $a \in \{0, 1\}$, whose 2×2 block in position (x, y) is

$$K_t[x, y] = \psi(\chi(y - x - t)).$$

Lemma 1. *For every $t \in \mathbb{F}$, $K_t K_t^T = 2(q+1)I_{2q} - 2\mathcal{R}$.*

Proof. The (x, x') block of $K_t K_t^T$, with $d = x' - x$ and $z = y - x - t$, equals $\sum_z \psi(\chi(z))\psi(\chi(z-d))^T$. For $d = 0$ each summand is $2I_2$, giving $2qI_2$. For $d \neq 0$ the terms $z = 0$ and $z = d$ carry opposite skew parts and cancel, using $\chi(-d) = \chi(d)$ (here $\chi(-1) = 1$); for $z \neq 0, d$, $\psi(\chi(z))\psi(\chi(z-d))^T = 2\chi(z)\chi(z-d)I_2$, and $\sum_z \chi(z(z-d)) = -1$ yields $-2I_2$. These are the diagonal and off-diagonal blocks of $2(q+1)I - 2\mathcal{R}$. $\square$

Lemma 2. *For $r, s \in \mathbb{F}$,*

$$\sum_{a \in \mathbb{F}} K_{ar} K_{as}^T = \begin{cases} 2q(q+1)I_{2q} - 2q\mathcal{R}, & r = s, \\ 2\mathcal{R}, & r \neq s. \end{cases}$$

Proof. The case $r = s$ is q copies of Lemma 1. For $r \neq s$, in the (x, x') block put $z = y - x - ar$; the second argument becomes $z - (x' - x) + a(r - s)$, which ranges over all of $\mathbb{F}$ independently of z as a does. By (2.1) each block is $ZZ^T = 2I_2$, i.e. the corresponding block of $2\mathcal{R}$. $\square$

For $r \in \mathbb{F}$ let B_r be the $2q \times 2q$ matrix whose row (x, a) equals row (r, a) of K_0 for every x ; thus B_r has only two distinct rows. From Lemma 1, $B_r B_s^T = 2q\mathcal{R}$ if $r = s$ and $-2\mathcal{R}$ if $r \neq s$.

Form the $2q \times 2q(q+1)$ block rows $M_r = [B_r \mid K_{ar} \ (a \in \mathbb{F})]$.

Lemma 3. $M_r M_s^T = 2q(q+1)I_{2q}$ if $r = s$, and 0 if $r \neq s$.

Proof. $M_r M_s^T = B_r B_s^T + \sum_a K_{ar} K_{as}^T$. For $r = s$ the $2q\mathcal{R}$ of the border cancels the $-2q\mathcal{R}$ of Lemma 2; for $r \neq s$ the $-2\mathcal{R}$ cancels the $2\mathcal{R}$. $\square$

Let C be the Paley conference matrix indexed by $\{\infty\} \cup \mathbb{F}$, namely $C_{u,u} = 0$, $C_{\infty,x} = C_{x,\infty} = 1$, $C_{x,y} = \chi(y - x)$, and let S be the order- $2(q+1)$ Paley-II Hadamard matrix obtained by replacing $0, 1, -1$ by $Z, P, -P$. Delete the two rows of S indexed by ∞ ; in each remaining row repeat the ∞ -column pair q times unchanged, and replace each finite column pair by its product with R , repeated q times. This is a $2q \times 2q(q+1)$ matrix M_∞ .

Lemma 4. $M_\infty M_\infty^T = 2q(q+1)I_{2q}$ and $M_\infty M_r^T = 0$ for all $r \in \mathbb{F}$.

Proof. The first identity holds because S is Hadamard, each column pair is repeated q times, and right multiplication by R preserves row inner products. For the cross identity, summing a cap row against a row of M_r and using (2.1) on every $2q$ -block leaves $(P_\alpha + (\sum_a S[e, a]_\alpha)R) \cdot Z_\beta$; the inner sum of finite entries of a row of S is Z_α , and $P + ZR = P - P = 0$. $\square$

Theorem 1. *Let $q \equiv 1 \pmod{4}$ be a prime power. Then*

$$H = \begin{bmatrix} M_\infty \\ M_r \ (r \in \mathbb{F}) \end{bmatrix}$$

is a Hadamard matrix of order $2q(q+1)$.

Proof. H is $\{1, -1\}$ and square of order $2q + q \cdot 2q = 2q(q+1)$. Lemmas 3 and 4 give $HH^T = 2q(q+1)I$. $\square$

Corollary 1. *For every prime power $q \equiv 1 \pmod{4}$ there is a Hadamard matrix of order $2q(q+1)$; the smallest non-prime instance is $q = 9$, of order 180.*

The construction has been verified by full Gram computation for all $q \leq 49$ (orders 60, $\dots$, 4900, including the extension fields $q = 9, 25, 49$) and by streamed row-orthogonality checks for $q = 81$ and $q = 125$ (orders 13284 and 31500).

4 Relation to the Farouk–Wang construction

Theorem 1 gives, for the $q \equiv 1 \pmod{4}$ case, an explicit form of the Scarpis-type family that Đoković [11] asked for. That family was already constructed by Farouk and Wang [1, 2] from the same order- $2(q+1)$ Paley-II seed. This section records how the two routes differ and where the orders of the family sit relative to the audited tables of known Hadamard orders.

While Theorem 1 and [1, Cor. 2.1] assert the same existence result, the routes differ. Farouk and Wang obtain the order- $2q(q+1)$ matrix from a lifting procedure applied to a Hadamard matrix of order $2(q+1)$ satisfying the eligibility conditions of [1, Lem. 2.2] — a formally larger input class,

with Paley II the only exhibited member — and verify orthogonality row pair by row pair through counting arguments (row sums, balanced sign-pair tallies, and a unique-intersection computation), importing the key inner-product values from the Hadamard property of the input matrix. The present note fixes the Paley-II seed and writes the finite bands entrywise in the closed form $\psi(\chi(y - x - t))$; the Hadamard property follows from four block-Gram identities in which the corresponding quantities are computed directly from the character sums (2.1) and $\sum_z \chi(z(z - d)) = -1$, together with $P + ZR = 0$ and the Hadamard property of the Paley-II seed S (itself a character-sum identity). The correspondence is exact: their unique-intersection index matches the affine collision $x + ar = x' + as$ in the substitution step of Lemma 2, and their imported inner products match the values that Lemma 1 derives from $\sum_z \chi(z(z - d)) = -1$. The devices used for the cap band also differ: a column permutation arranging balanced sign pairs in [1] versus right multiplication by R here. The two constructions moreover produce genuinely different matrices, not merely different presentations: at $q = 5$ the matrix of Theorem 1 is not Hadamard-equivalent to the order-60 example printed in the appendix of [1] — the multiset of absolute row-quadruple correlations $|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}|$, an invariant of Hadamard equivalence, differs for the two matrices (Figure 1). Since Farouk and Wang note that their output may vary with the choice of bijection α , we leave open whether the matrix of Theorem 1 is equivalent to some other output of their procedure.

Since $q \equiv 1 \pmod{4}$, the 2-adic valuation of the order is exactly 2:

$$2q(q + 1) = 4 \cdot q \cdot \frac{q+1}{2}, \quad q \text{ and } \frac{q+1}{2} \text{ both odd.} \quad (4.1)$$

Thus every order is $4 \times (\text{odd})$, so it is never a Kronecker product of two smaller Hadamard matrices, and its odd part $q(q + 1)/2$ is always composite. Two consequences follow. First, the family never meets the dominant list of open Hadamard orders, which are $4 \times \text{prime}$. Second, (4.1) has the form needed for Turyn’s construction: when T -matrices of order t and Williamson-type matrices of order w are available, one obtains a Hadamard matrix of order $4tw$ [7, 8], and here $\{t, w\} = \{q, (q + 1)/2\}$. The coverage analysis of Appendix A makes both points quantitative: all orders with odd part ≤ 3000 are recorded in the standard tables, the first order beyond that range is covered by such a Turyn product, and a restricted screen beyond the audited range records exactly which orders it does and does not cover. Failure of that screen is not a claim that an order is open.

A Coverage analysis

Write $N = 2q(q + 1)$ and $m = q(q + 1)/2$ for its odd part, so $N = 4m$. We assess coverage of N by prior constructions on three levels: the audited tables (§A.1), the first unaudited order (§A.2), and an elementary-witness test that isolates candidate orders (§A.3).

A.1 Orders with odd part at most 3000

The surveys of Seberry and Yamada [8, 13] and the database of Cati and Pasechnik [14] record, for every odd $m \leq 3000$, the least t for which a Hadamard matrix of order $2^t m$ is known. The default is $t = 2$ (so $N = 4m$ is known), and the only odd $m \leq 3000$ with $t > 2$ are the primes 167, 179, 223, 283 (orders 668, 716, 892, 1132, still open). Since $m = q(q + 1)/2$ is composite it is never one of these; hence *every N with $m \leq 3000$ is a known Hadamard order*. This is the range $q \leq 73$, tabulated below with an explicit elementary witness where one exists (Paley I/II, or a Turyn product $4tw$).

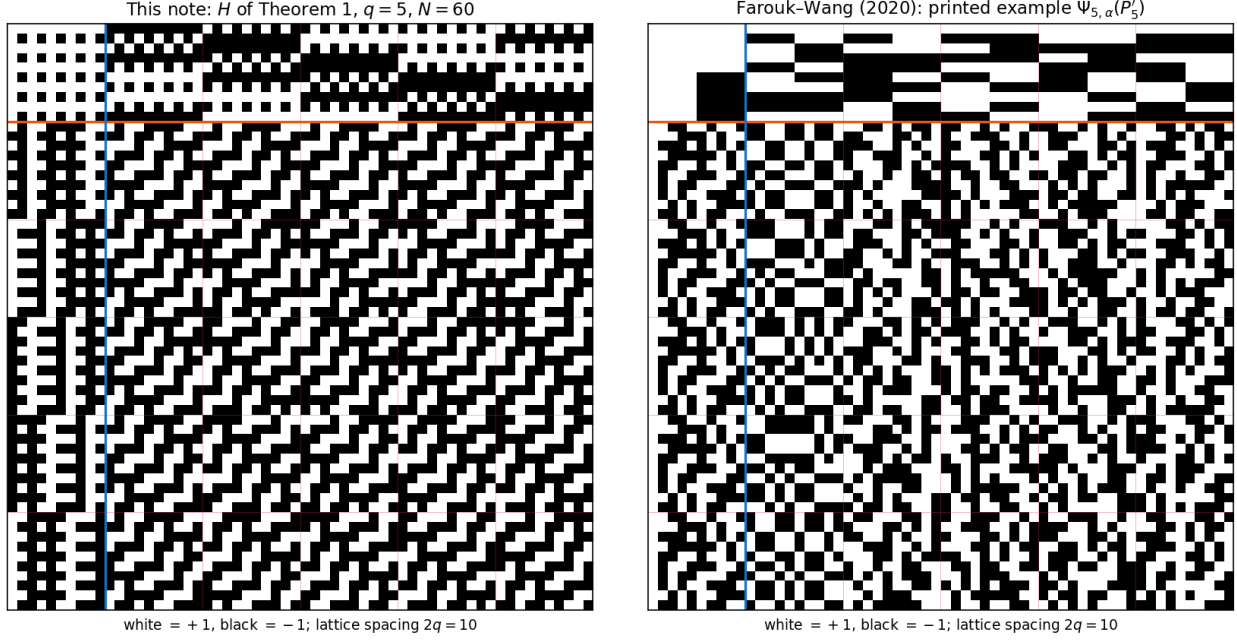


Figure 1: The two constructions at $q = 5$ (order 60; white = +1, black = -1; lattice spacing $2q = 10$). Left: the matrix of Theorem 1 — the cap band M_∞ above the orange separator is a family of period-two column patterns, and the finite bands M_r carry the diagonal texture of the shifts $\psi(\chi(y - x - ar))$. Right: the order-60 example $\Psi_{5,\alpha}(P'_5)$ printed in the appendix of [1], whose top band is a coarse tiling (its border block consists of four constant 5×5 quadrants) and whose finite bands show no diagonal structure. The two matrices are Hadamard-inequivalent: over the $\binom{60}{4} = 487,635$ row quadruples, the right matrix attains $|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}| = 28$ on 400 quadruples, the left on none, and this multiset is invariant under row and column permutations and negations (and the comparison fails in both orientations, covering the transpose convention as well). This separates the printed representative only: outputs of the Farouk–Wang procedure for other choices of α are not addressed.

q	$N = 2q(q + 1)$	odd part m	witnessing construction
5	60	$3 \cdot 5$	Paley II, $p = 29$
9	180	$3^2 \cdot 5$	Paley II, $p = 89$
13	364	$7 \cdot 13$	Paley II, $p = 181$
17	612	$3^2 \cdot 17$	Turyn $4 \cdot 9 \cdot 17$
25	1300	$5^2 \cdot 13$	Turyn $4 \cdot 13 \cdot 25$
29	1740	$3 \cdot 5 \cdot 29$	Turyn $4 \cdot 15 \cdot 29$
37	2812	$19 \cdot 37$	Turyn $4 \cdot 19 \cdot 37$
41	3444	$3 \cdot 7 \cdot 41$	Paley II, $p = 1721$
49	4900	$5^2 \cdot 7^2$	Turyn $4 \cdot 25 \cdot 49$
53	5724	$3^3 \cdot 53$	Paley II, $p = 2861$
61	7564	$31 \cdot 61$	Turyn $4 \cdot 31 \cdot 61$
73	10804	$37 \cdot 73$	tables; no restricted witness (see §A.3)

The entry $q = 73$ is instructive: the restricted test below supplies no divisor-split witness for its odd part $2701 = 37 \cdot 73$, yet $N = 10804$ is a known Hadamard order because $m \leq 3000$ is audited. Failure of the elementary test is therefore a property of that test, not of Hadamard existence.

A.2 The first unaudited order: $q = 81$

The smallest order of the family whose odd part exceeds 3000, and hence lies beyond the audited range, is

$$q = 81 = 3^4, \quad N = 13284, \quad m = 3321 = 3^4 \cdot 41 > 3000.$$

It is nevertheless a known Hadamard order. By (4.1),

$$13284 = 4 \cdot 41 \cdot 81 = 4tw, \quad t = 41, \quad w = 81,$$

and both ingredients are classical: T -matrices of order 41 come from the base sequences $BS(21, 20)$ [7, 12], and Williamson-type matrices of order $81 = 9^2$ come from Turyn's 9-power family [7, 8]. The Turyn array on these [8, 13] yields a Hadamard matrix of order $4 \cdot 41 \cdot 81 = 13284$. Thus $q = 81$ is the smallest unaudited order, and it is covered.

A.3 An elementary-witness test and the candidate orders

Beyond the audited range we have no table to appeal to, so we apply a deliberately conservative *elementary-witness* test based on (4.1) and Turyn's array. We call an odd order t *T-known* when $t \leq 59$; the required T -matrices follow from the known base sequences $BS((t+1)/2, (t-1)/2)$, whose second parameter is at most 29. We call an odd order w *W-known* when $w \leq 33$, or when $2w-1$ is a prime power; in the latter case Turyn's $(P+1)/2$ family, with $P = 2w-1 \equiv 1 \pmod{4}$, supplies the required Williamson-type matrices. An unaudited order $N = 4m$ passes the test if N is a Paley I/II order, or if there is an *exhibited* factorization

$$m = tw, \quad t \text{ T-known}, \quad w \text{ W-known}. \quad (\text{C.1})$$

Thus every passing verdict is an unconditional construction from the ingredients just listed; no multiplicative closure is assumed.

Remark 1. *An earlier version of this appendix asserted that classes of eligible T - and Williamson orders were multiplicatively closed and then classified m prime factor by prime factor. That assertion was uncited, and we withdraw both it and the factorwise test. Even for Williamson matrices proper, unrestricted closure is false: Đoković [9] enumerated Williamson matrices at orders 33 and 39, but proved that none exist at $35 = 5 \cdot 7$, although they exist at both orders 5 and 7. Holzmänn, Kharaghani and Tayfeh-Rezaie [10] later found the further nonexistence orders 47, 53, and 59. Terminology in the literature includes broader variants called Williamson-type matrices; we make no closure claim for any such class. Instead, (C.1) requires the two actual ingredient orders used by Turyn's array to be known individually.*

The factorwise test was unreliable in both directions. It could accept an m merely because each prime factor was eligible, without producing a pair t, w . It could also reject an m because of a prime factor that nevertheless lies inside a directly known composite W -order. For example, at $q = 197$,

$$m = 19503 = 3 \cdot 6501, \quad 2 \cdot 6501 - 1 = 13001$$

is prime, so $t = 3, w = 6501$ is a witness under (C.1); the old factorwise test rejected it because of the prime factor 197. This is why the table below is recomputed from actual divisor splits rather than inherited from the earlier screen.

The audited range of §A.1 is a table lookup, and the $q = 81$ witness of §A.2 cites its two ingredients individually. Both are independent of this correction. Nothing in §§1–4, and in particular neither Theorem 1 nor its proof, depends on this appendix.

A passing verdict now exhibits a construction. A failing verdict (*no restricted elementary witness*) is a statement about this test only, not a claim that N is open. Over the 232 orders with $q \leq 3000$, the strict test covers 128 (audited, Paley, or Turyn) and leaves 104 without a witness. The smallest failure remains

$$q = 109, \quad N = 23980, \quad m = 5 \cdot 11 \cdot 109,$$

for which none of the possible T -known divisors produces a W -known complementary factor. In particular, $2 \cdot 109 - 1 = 217 = 7 \cdot 31$ is not a prime power. This $N = 23980$ is the smallest order of the family for which we exhibit neither an audited entry nor a restricted elementary witness; we do not assert it is open. Williamson-type matrices of order 109, if available, would settle it at once using $t = 55, w = 109$ (compare $q = 73$).

Table 1 lists all 104 failures of this restricted test for $q \leq 3000$. It is a reproducible classification relative to the stated ingredient sets, not a literature-complete classification of the corresponding Hadamard orders.

Table 1: Orders $N = 2q(q + 1)$, q a prime power $\equiv 1 \pmod{4}$, with no restricted elementary witness (§A.3), $109 \leq q \leq 2969$.

q	N	odd part m
109	23980	$5 \cdot 11 \cdot 109$
121	29524	$11^2 \cdot 61$
157	49612	$79 \cdot 157$
173	60204	$3 \cdot 29 \cdot 173$
229	105340	$5 \cdot 23 \cdot 229$
277	154012	$139 \cdot 277$
289	167620	$5 \cdot 17^2 \cdot 29$
313	196564	$157 \cdot 313$
397	316012	$199 \cdot 397$
409	335380	$5 \cdot 41 \cdot 409$
421	355324	$211 \cdot 421$
457	418612	$229 \cdot 457$
529	560740	$5 \cdot 23^2 \cdot 53$
577	667012	$17^2 \cdot 577$
601	723604	$7 \cdot 43 \cdot 601$
613	752764	$307 \cdot 613$
617	762612	$3 \cdot 103 \cdot 617$
641	823044	$3 \cdot 107 \cdot 641$
661	875164	$331 \cdot 661$
733	1076044	$367 \cdot 733$
757	1147612	$379 \cdot 757$
797	1272012	$3 \cdot 7 \cdot 19 \cdot 797$
821	1349724	$3 \cdot 137 \cdot 821$
829	1376140	$5 \cdot 83 \cdot 829$
841	1416244	$29^2 \cdot 421$
853	1456924	$7 \cdot 61 \cdot 853$
877	1540012	$439 \cdot 877$
937	1757812	$7 \cdot 67 \cdot 937$
961	1848964	$13 \cdot 31^2 \cdot 37$
997	1990012	$499 \cdot 997$
1009	2038180	$5 \cdot 101 \cdot 1009$
1021	2086924	$7 \cdot 73 \cdot 1021$
1033	2136244	$11 \cdot 47 \cdot 1033$
1097	2409012	$3^2 \cdot 61 \cdot 1097$
1109	2461980	$3 \cdot 5 \cdot 37 \cdot 1109$
1117	2497612	$13 \cdot 43 \cdot 1117$
1129	2551540	$5 \cdot 113 \cdot 1129$

q	N	odd part m
1153	2661124	$577 \cdot 1153$
1181	2791884	$3 \cdot 197 \cdot 1181$
1201	2887204	$601 \cdot 1201$
1213	2945164	$607 \cdot 1213$
1237	3062812	$619 \cdot 1237$
1249	3122500	$5^4 \cdot 1249$
1297	3367012	$11 \cdot 59 \cdot 1297$
1321	3492724	$661 \cdot 1321$
1361	3707364	$3 \cdot 227 \cdot 1361$
1373	3773004	$3 \cdot 229 \cdot 1373$
1381	3817084	$691 \cdot 1381$
1453	4225324	$727 \cdot 1453$
1481	4389684	$3 \cdot 13 \cdot 19 \cdot 1481$
1553	4826724	$3 \cdot 7 \cdot 37 \cdot 1553$
1597	5104012	$17 \cdot 47 \cdot 1597$
1601	5129604	$3^2 \cdot 89 \cdot 1601$
1609	5180980	$5 \cdot 7 \cdot 23 \cdot 1609$
1637	5362812	$3^2 \cdot 7 \cdot 13 \cdot 1637$
1657	5494612	$829 \cdot 1657$
1681	5654884	$29^2 \cdot 41^2$
1709	5844780	$3^2 \cdot 5 \cdot 19 \cdot 1709$
1753	6149524	$877 \cdot 1753$
1777	6319012	$7 \cdot 127 \cdot 1777$
1789	6404620	$5 \cdot 179 \cdot 1789$
1873	7020004	$937 \cdot 1873$
1877	7050012	$3 \cdot 313 \cdot 1877$
1933	7476844	$967 \cdot 1933$
2017	8140612	$1009 \cdot 2017$
2029	8237740	$5 \cdot 7 \cdot 29 \cdot 2029$
2053	8433724	$13 \cdot 79 \cdot 2053$
2089	8732020	$5 \cdot 11 \cdot 19 \cdot 2089$
2113	8933764	$7 \cdot 151 \cdot 2113$
2137	9137812	$1069 \cdot 2137$
2161	9344164	$23 \cdot 47 \cdot 2161$
2197	9658012	$7 \cdot 13^3 \cdot 157$
2209	9763780	$5 \cdot 13 \cdot 17 \cdot 47^2$
2213	9799164	$3^3 \cdot 41 \cdot 2213$
2269	10301260	$5 \cdot 227 \cdot 2269$
2273	10337604	$3 \cdot 379 \cdot 2273$
2281	10410484	$7 \cdot 163 \cdot 2281$
2297	10557012	$3 \cdot 383 \cdot 2297$
2341	10965244	$1171 \cdot 2341$
2377	11305012	$29 \cdot 41 \cdot 2377$
2381	11343084	$3 \cdot 397 \cdot 2381$
2389	11419420	$5 \cdot 239 \cdot 2389$
2401	11534404	$7^4 \cdot 1201$
2417	11688612	$3 \cdot 13 \cdot 31 \cdot 2417$
2437	11882812	$23 \cdot 53 \cdot 2437$
2473	12236404	$1237 \cdot 2473$
2557	13081612	$1279 \cdot 2557$
2617	13702612	$7 \cdot 11 \cdot 17 \cdot 2617$
2621	13744524	$3 \cdot 19 \cdot 23 \cdot 2621$
2657	14124612	$3 \cdot 443 \cdot 2657$
2689	14466820	$5 \cdot 269 \cdot 2689$
2693	14509884	$3 \cdot 449 \cdot 2693$
2713	14726164	$23 \cdot 59 \cdot 2713$
2749	15119500	$5^3 \cdot 11 \cdot 2749$
2777	15429012	$3 \cdot 463 \cdot 2777$

q	N	odd part m
2797	15652012	$1399 \cdot 2797$
2801	15696804	$3 \cdot 467 \cdot 2801$
2833	16057444	$13 \cdot 109 \cdot 2833$
2837	16102812	$3 \cdot 11 \cdot 43 \cdot 2837$
2857	16330612	$1429 \cdot 2857$
2909	16930380	$3 \cdot 5 \cdot 97 \cdot 2909$
2917	17023612	$1459 \cdot 2917$
2953	17446324	$7 \cdot 211 \cdot 2953$
2969	17635860	$3^3 \cdot 5 \cdot 11 \cdot 2969$

Every odd part in Table 1 is composite by (4.1), so these orders do not meet the $4 \times$ prime form that dominates the initial list of open Hadamard orders. At the larger sizes in the table we make no literature-complete assertion of known or open status. Widening the ingredient set (for example with computer-tabulated Williamson and T -matrices or Goethals–Seidel cyclotomic difference sets) can remove entries from this restricted list. The role of Theorem 1 is to give a single closed-form rule for the whole family, including the region this elementary screen does not reach, rather than to derive novelty from a coverage-table gap.

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